Darren Biskup

408-680-7370 | biskupdarren@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Columbia University — New York, NY
M.S./Ph.D. Mechanical Engineering (Robotics & Rehabilitation) — Aug 2024 - Present
Columbia-Dream Sports AI Ph.D. Fellowship — GPA: 4.11/4.33

University of Illinois, Urbana-Champaign — Urbana, IL
B.S. Mechanical Engineering, Computer Science Minor — May 2024
High Honors, James Scholar Honors Program, Dean's List — GPA: 3.95

SKILLS

Languages & Tools: C/C++, Python (PyTorch, JAX, OpenCV), C#/Unity, Linux/Bash, Git, CMake
Controls & Estimation: Convex MPC, Trajectory Optimization, Sensor Fusion (EKF/UKF), ALGLIB, OSQP
Autonomy & Simulation: MuJoCo/MJPC, pyElastica, ROS2, Predictive Safety Filters, Domain Randomization
Hardware & Systems: CAN/CAN-FD, BLDC control, Raspberry Pi, pneumatics, force plates/load cells

EXPERIENCE

Robotics And Rehabilitation (ROAR) Lab, Columbia University — Aug 2024 - Present
Graduate Research Assistant

- Authored an IEEE RA-L paper on real-time ground-reaction-force-aware convex MPC using human centroidal dynamics for underactuated cable-driven robotic standing assistance. [Paper]
- Implemented the MPC as a C#/Unity QP using ALGLIB QuickQP, reducing dynamic tracking error by 24% and mechanical work by 29% versus impedance control; [Code].
- Integrated a hierarchical real-time control stack with host-level MPC, asynchronous TCP tension-command streaming, 1-kHz load-cell feedback, and CAN-controlled EPOS motor drives, incorporating Vicon/HTC Vive motion capture and force-plate sensing.

Mehta Research Group/GazzolaLab, UIUC — Jan 2023 - May 2024
Undergraduate Research Assistant - Soft Robotics

- Built an OpenCV/scikit-image calibration pipeline to extract accurate 3D marker coordinates from a custom spatial grid for robotic-arm pose tracking.
- Developed Python control scripts synchronizing a physical joystick-actuated soft robotic arm with its *pyElastica* digital twin to evaluate sim-to-real dynamic fidelity.

Skydio — May 2022 - Aug 2022
Product Design Engineering Intern

- Designed navigation-camera mounting features for the Skydio X10 and coordinated with overseas vendors to implement design changes while meeting injection-molding and lead-time constraints.
- Collaborated with manufacturing and autonomy teams on design-for-manufacturing (DFM) and spatial hardware layout for the next-generation autonomous performance quadcopter.

Lucid Motors — May 2021 - Aug 2021
Mechanical Engineering Intern, High Voltage

- Conducted data-driven thermal and efficiency analysis on high-voltage battery pack assemblies to optimize hardware performance and range.

ROBOTICS & AUTONOMY PROJECTS

Predictive Safety Filter for MuJoCo Quadrotor Flight — *Python, JAX, MuJoCo, SciPy*

- Implemented a finite-horizon predictive safety filter for online collision avoidance of a MuJoCo quadrotor in constrained workspaces with modeled obstacles. Optimizes motor commands over forward-simulated trajectories to minimize deviation from requested inputs while maintaining collision-free motion.
- Used JAX's just-in-time (JIT) compilation and automatic differentiation to accelerate repeated nonlinear physics rollouts and SLSQP cost and constraint evaluations.

C++ MuJoCo GRF Data-Generation Pipeline — *MuJoCo, MJPC, C++, CMake*

- Building a C++17 MuJoCo/MJPC data factory that replays motion-capture-derived anatomical marker trajectories and records per-foot GRFs, contact diagnostics, and replayable runs.
- Implementing quasi-random morphology and marker-layout sampling across male and female humanoid models to study tracking and contact robustness.
- Added reproducible CMake builds plus contract and integration tests for simulation and run-bundle invariants.